

A Profile of Adolescent-Onset Epilepsy

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To describe the profile of children with adolescent-onset epilepsy and to determine factors predictive of outcome. A database was searched for all patients with a first seizure between the age of 12 and 16 years. Sixty-five adolescents met inclusion criteria. Ten patients needed at least two medications to control seizures, 36 remained on medication at their last visit and 12 patients had at least 1 seizure in the year preceding. A diagnosis of juvenile myoclonic epilepsy, the presence of coexisting seizures, coexisting myoclonic seizures, age

≤14.5 years at initial diagnosis, and the presence of compliance issues were significantly associated with the need for medication at last visit. Female gender and the presence of compliance issues were associated with the occurrence of at least 1 seizure in the year preceding last visit. A good outcome for adolescent-onset epilepsy can generally be expected in the short term.

Keywords: adolescence; epilepsy; outcomes; seizures

Adolescence, which is derived from the Latin *adolescere* meaning to come to maturity, represents a period of major change across a variety of personal and social domains. Structurally, the brain continues to gain volume until the age of 14 years, whereas white matter volume increases until the early 20s.¹ The psychological profile of adolescence also has its impact on the health status of this population, where the constant search for independence and autonomy often interacts with efforts to control chronic diseases like epilepsy, which represents the most common neurological disorder of adolescence. In view of this very unique period of life, some studies have already proposed models of “transition clinics,” addressing the special needs of adolescents with epilepsy.²⁻⁴ To provide good care to this population, one also needs an understanding of the profile and natural history of adolescent-onset epilepsies considered as a group not only in terms of specific epileptic syndromes.

Experts in the field of pediatric epilepsy consider that the adolescent-onset epilepsies often have a rather benign

course, with most being self-limited, with a low incidence of drug resistance. Neurological and psychiatric comorbidities are also traditionally considered rare in this population, perhaps reflecting the often nonlesional etiology of these epilepsies.⁵

The recent literature has been largely directed to specific epileptic syndromes, including the well-recognized idiopathic generalized epilepsies like juvenile myoclonic epilepsy.^{6,7} Unfortunately, not all adolescents with new-onset epilepsy fall into a recognized epileptic syndrome, which explains the great need to clarify the profile of adolescent-onset epilepsy as a group. Moreover, the identification of possible risk factors for drug resistance or for a poor outcome could help in the early identification of adolescents at higher risk and thus allow the introduction of more aggressive and targeted early interventions. It is also reasonable to think that this early identification and intervention could ameliorate some of the adverse social, educational, and personal consequences related to adolescent-onset epilepsy. In this context, the objectives of this study were to describe the profile of children with adolescent-onset epilepsy and to determine factors predictive of short-term outcome.

Methods

Participants were identified via the standardized computerized database of a single pediatric neurologist (M.S.). Patients entered into this database were examined by the neurologist in 3 different clinical settings: (a) hospital-based ambulatory clinic, (b) hospital-based private neurology office (1991-1996), and (c) suburban private pediatric neurology clinic. The database contained 14 domains pertaining to demographic, diagnostic, and treatment windows with information entered at initial visit and updated at each follow-up visit. The database was systematically

Received January 14, 2009. Received revised February 13, 2009. Accepted for publication February 13, 2009.

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The authors have no conflicts of interest to disclose with regard to this article. MS is grateful for the support of the MCH Foundation during the writing of this manuscript. Alba Rinaldi provided the necessary secretarial assistance.

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Simard-Tremblay E, Shevell M. A profile of adolescent-onset epilepsy. *J Child Neurol*. 2009;24:1243-1249.

searched for all patients with an onset of epilepsy between the age of 12 and 16 years. The medical charts of the patients previously identified were then reviewed by one investigator (E.S.-T.).

All patients meeting the following inclusion criteria and none of the exclusion criteria were included in the study. The inclusion criteria consist of (a) initial intake and neurologic assessment for a suspected seizure between July 1, 1991, and June 30, 2006, (b) first afebrile seizure between 12 and 16 years, (c) presumptive diagnosis of epilepsy (at least 2 unprovoked seizures or treatment after the first seizure), and (d) minimum follow-up period of 24 months. All patients seen in the context of a second opinion were excluded from the study.

The following information was obtained through systematic inpatient and outpatient chart review; gender, family history of epilepsy or febrile seizure, perinatal history, prior febrile seizure, comorbid conditions, seizure type at presentation, age at first suspected seizure, number of seizures prior to diagnosis, coexisting seizures, neurologic examination, most abnormal electroencephalogram (EEG) before initiation of treatment, neuroimaging, treatment, presence of an epileptic syndrome as defined by the International League Against Epilepsy,⁸ and the clinical status at last visit.

A normal perinatal history was defined as a term delivery without evident asphyxia or malformation and with no need for neonatal intensive care unit admission other than for routine observation. As comorbid conditions, the presence of mental retardation, attention-deficit hyperactivity disorder, learning disability, tics, Tourette syndrome, behavioral problems (ie, conduct disorder, oppositional defiance disorder, episodic dyscontrol syndrome), and affective and psychotic disorders were recorded. Those disorders were all defined by their respective *Diagnostic and Statistical Manual of Mental Disorders* (Fourth Edition) criteria. The following characteristics of the neurologic examination were noted: microcephaly, macrocephaly, and focal neurological signs. All the EEGs performed before initiation of treatment were reviewed for the presence of an abnormal background (focal or generalized), epileptiform activity (focal, multifocal, and generalized), and any evident triggers of epileptiform activity (sleep, hyperventilation, and photic stimulation). The most abnormal of these EEGs was then retained for analysis. In the rare case where no EEG was available before initiation of treatment, EEGs performed within 1 month of initiation of treatment were searched for the same features.

The treatment of epilepsy was recorded as the number and specific type of medication needed to achieve effective seizure control and documented need for ketogenic diet, vagal nerve stimulator, or epilepsy surgery. Medications stopped for an allergic reaction or side effects were not considered in the analysis. The age of the patient at last visit along with the time interval between first and last visit was recorded. It was also noted if the patient at the time of the last clinic visit was still on antiepileptic medication, the level of seizure control (excellent, moderate, poor), and the disposition of the patient at last visit (ie, continue follow-up in pediatric neurology, tapered off prior medications, transfer to adult care). Excellent control was defined as no evident seizure in the prior year while on medication, whereas moderate or poor seizure control were respectively defined as 1 to 5 seizures and more than 6 seizures in the year preceding the last clinic visit while on medication.

Table 1. Group Profile

	Frequency	Percentage
Total	65	
Males	35	53.8
Females	30	46.2
Family history of epilepsy ^a	26	41.3
First-degree relative affected	4	6.3
Second- or third-degree relative affected	19	30.2
Both first- and second- or third-degree relative affected	3	4.8
Family history of febrile seizure ^a	2	3.2
Perinatal history abnormal ^b	8	12.3
Preeclampsia	2	3.1
Prematurity	2	3.1
Pneumothorax	1	1.6
Asphyxia	1	1.6
Urgent C-section for fetal distress	1	1.6
Neonatal intensive care unit admission	3	4.7
>1 perinatal risk factor	2	3.1
Prior febrile seizure	7	10.8
Typical febrile seizure	5	7.7
Atypical febrile seizure	2	3.1
Abnormal development	1	1.5
Age out of typical range	1	1.5
Comorbid psychiatric conditions at presentation	19	29.2
Mental retardation	4	6.2
Attention-deficit hyperactivity disorder (ADHD)	3	4.6
Learning disorder	10	15.4
Affective disorder	2	3.1
Anxiety disorder	2	3.1
Psychotic disorder	1	1.5
>1 comorbid psychiatric condition	3	4.6
Physical examination abnormal at presentation	6	9.2
Macrocephaly	2	3.1
Focal findings	4	6.2

a. Information available for 63 patients.

b. Information available for 64 patients.

Statistical analysis was performed using SPSS Statistics (Version 17.0) software. The profile of children needing only 1 medication (group 1) and of children needing at least 2 medications to control their seizures (group 2) were then compared using the Pearson χ^2 , Fisher exact, and Student *t* tests with a *P* value of .05 or less established a priori as the threshold for statistical significance. The same method was used to try to identify any factor predicting if the patient remained on an anti-epileptic drug and the level of control of seizures documented at the last clinic visit.

Results

Descriptive Statistics

Sixty-five patients met all the inclusion and none of the exclusion criteria (Table 1). Of those, 35 (53.8%) were males and 30 (46.2%) were females. Family history was available for 63 patients and was positive for a first-degree

Table 2. Epilepsy Profile

	Frequency	Percentage
Seizure at presentation		
Generalized	38	58.5
Myoclonic	3	4.6
Typical absence	4	6.2
Tonic-clonic (GTC)	31	47.7
Partial	27	41.5
Complex partial	8	12.3
Secondary generalization	19	29.2
Age at first suspected seizure		
Mean: 14.0 years (SD $\pm$ 1.4 years)		
Range: 12-16.7 years		
Coexisting seizures	20	30.8
Myoclonic seizures with GTC and/or typical absences	11	16.9
Combination of GTC and typical absences	2	3.1
Simple and/or complex partial and/or secondary generalized seizures	7	10.8
Abnormal EEG at presentation ^a	53	85.5
Abnormal background, no epileptic activity	9	17.0
Presence of epileptic activity with normal background	37	69.8
Abnormal background and presence of epileptic activity	7	13.2
Neuroimaging performed	58	89.2
CT scan only	39	60.0
CT scan and MRI	19	29.2
Neuroimaging abnormal	9	15.5
CT scan only	2	3.4
MRI only	2	3.4
CT scan and MRI	5	8.6
Number of AED needed to achieve seizure control		
One AED	55	84.6
Two AED	7	10.8
Three AED	2	3.1
Five AED	1	1.5
Epilepsy surgery	2	3.1
Tumor removal	1	1.5
Granuloma removal (Cysticercosis)	1	1.5

Note: AED, antiepileptic drug; CT, computed tomography; EEG, electroencephalogram; GTC, generalized tonic-clonic; MRI, magnetic resonance imaging.

a. EEG available for 62 patients.

relative with epilepsy in 7 cases (11.1%) and for a second- or third-degree relative with epilepsy in 22 cases (34.9%). The neonatal period was considered normal in the majority of patients, with only 8 patients (12.3%) having an abnormal neonatal period. Seven patients (10.8%) had a history of at least 1 febrile seizure, with 5 (7.7%) described as a simple febrile seizure. At least 1 psychiatric comorbid condition was present at presentation in 19 patients (29.2%), with 3 (4.6%) having 2 psychiatric or neurologic comorbid conditions. Learning difficulties were present in 10 adolescents, representing 15.4% of the clinical sample.

The seizure at initial presentation was generalized in 58.5% ($n = 38$), and the mean age of occurrence of the first suspected seizure was 14.0 years (SD $\pm$ 1.4 years; Table 2). In 30.8% ($n = 20$) of cases, patients had

Table 3. Epileptic Syndromes (ILAE)

	Frequency	Percentage
Generalized		
Idiopathic generalized epilepsy syndrome	39	60
Juvenile absence epilepsy	2	3.1
Juvenile myoclonic epilepsy	13	20
Other generalized idiopathic epilepsies	24	36.9
Partial	26	40
Focal idiopathic epilepsy syndrome	4	6.2
Benign epilepsy of childhood with rolandic spikes	1	1.5
Benign occipital epilepsy	3	4.6
Cryptogenic focal epilepsy	16	24.6
Symptomatic focal epilepsy	6	9.2
Cerebral palsy	1	1.5
Posttraumatic epilepsy	1	1.5
Cysticercosis	1	1.5
Postencephalitis	1	1.5
Brain tumor	1	1.5
Cortical dysplasia	1	1.5

Note: ILAE, International League Against Epilepsy

experienced, at least once, another type of seizure. Eleven patients (16.9%) had a combination of myoclonic seizures with either generalized tonic-clonic or absence seizures, or both. Another 2 (3.1%) had experienced both generalized tonic-clonic and absence seizures, with no apparent history of myoclonic seizures and 7 (10.8%) had a combination of simple partial, complex partial, and secondary generalized seizures.

An EEG performed within the first month of initiation of treatment was available for 62 adolescents and was considered abnormal in 53 (85.5%). The abnormality consisted of an abnormal background without any epileptiform activity in 9 (17.0%), both an abnormal background and an epileptiform activity in 7 (13.2%) and epileptiform activity with a normal background in 37 (69.8%) of cases. Neuroimaging was performed in 89.2% of cases and was found to be abnormal in 9, representing 15.5% of those who underwent neuroimaging. Treatment was started in all adolescents with 2 documented seizures or at parental request after a first seizure with an epileptiform EEG (Table 3). In the overwhelming majority (84.6%), only 1 antiepileptic medication was needed to achieve eventual effective seizure control. Two adolescents had an epilepsy surgery performed, 1 for a tumor removal and then another for cysticercosis granuloma removal (ie, mass effect). No adolescents required the introduction of either the ketogenic diet or the insertion of a vagal nerve stimulator to achieve seizure control.

According to the criteria of the latest International League against Epilepsy classification of epilepsy syndromes,⁸ 39 adolescents (60%) were classified as having an idiopathic generalized epilepsy syndrome. Of these, 2 (3.1%) had juvenile absence epilepsy, 13 (20%) juvenile myoclonic epilepsy, and 24 (36.9%) were classified in the

Table 4. Outcome

	Frequency	Percentage
Mean follow-up period: 37.3 months (SD $\pm$ 12.8 months)		
Medication at last visit		
Still on antiepileptic medication	36	55.4
One medication	32	49.2
Two medications	3	4.6
Three medications	1	1.5
Off or tapering antiepileptic medication	29	44.6
Seizure control in prior year		
Excellent	53	81.5
Moderate	6	9.2
Poor	6	9.2
Compliance issue documented	12	18.5
Status/disposition at last visit		
Discharged from neurology	29	44.6
Continue care with pediatric neurology	6	9.2
Transferred to adult neurology	30	46.2

other generalized idiopathic epilepsies category. Four (6.1%) had a focal idiopathic epilepsy syndrome, 1 with benign epilepsy of childhood with rolandic spikes/benign epilepsy with centrotemporal spikes and 3 with benign occipital epilepsy. The focal epilepsy was lesional in 6 adolescents and 16 adolescents fell in the cryptogenic focal epilepsy group (Table 4).

At last visit, 36 adolescents (55.4%) remained on an antiepileptic medication, 32 of those being on only 1 drug (Table 5). Medication was discontinued usually after 2 years seizure free in the absence of a repeat actively epileptiform EEG. Control of seizures was excellent in 53 (81.5%), whereas 12 (18.5%) experienced at least 1 seizure in the previous year. Following the last appointment, 29 patients (44.6%) were discharged from neurology care and 36 patients (55.4%) either continued care with their pediatric neurologist ($n = 6$) or were transferred to an adult care setting ($n = 30$). Compliance issues were documented at least once during follow-up in 12 adolescents (18.5%).

Chi-Square and Fisher Exact Analysis

A diagnosis of juvenile myoclonic epilepsy ($P = .02$), the presence of coexisting seizures ($P = .008$), coexisting myoclonic seizures ($P = .03$), age ≤ 14.5 years at first seizure ($P = .01$), and the presence of a compliance issue documented ($P = .03$) were all associated with the need for continued antiepileptic drug at the last clinic visit. Female gender ($P = .03$) and the presence of a compliance issue ($P = .04$) were both associated with the occurrence of at least 1 seizure in the year preceding the last clinic visit (Table 6). No factor studied was found to be associated

Table 5. Chi-Square and Fisher Exact Analysis

	On Medication at Last Visit	Off Medication at Last Visit	Chi-Square Value	P Value
Age ≤ 14.5 years at first suspected seizure	27	13	6.178	.013
Diagnosis of Juvenile myoclonic epilepsy	11	2	5.619	.018
Coexisting seizures	16	4	7.084	.008
Coexisting myoclonic seizures	9	1	5.731	.034
Compliance issue documented	10	2	4.652	.031
	No Seizure in Previous Year	≥ 1 Seizure in Previous Year	Chi-Square Value	P Value
Female gender	21	9	4.928	.026
Compliance issue documented	7	5	5.264	.036

with the need for more than 1 antiepileptic drug for eventual seizure control.

Discussion

This study shows an overall good prognosis for adolescent-onset epilepsy. Indeed, almost 85% of the patients were successfully controlled on monotherapy, with 10.8%, 3.1%, and 1.5% being on 2, 3, and 4 medications, respectively. A previous study performed in adolescents with both childhood-onset and adolescence-onset epilepsy showed very similar results with 85% of the sample controlled on monotherapy.³ Moreover, in our study, almost 82% of the patients had excellent control of their seizures while on medication during the year preceding last follow-up. This overall good prognosis is concordant with the traditional thought that the great majority of adolescent-onset epilepsies responds well to an antiepileptic agent and thus would be compatible with a reasonably normal quality of life.⁵ Multiple studies performed in individuals with epilepsy of various ages have shown that a high seizure frequency and polypharmacy were strongly associated with increased psychosocial difficulties (eg, lower self-esteem, decreased quality of life, and higher perceived stigma), making the early identification of those children at higher risk of having a poor prognosis important.⁹⁻¹²

In this study, female gender and the presence of a compliance issue documented at any time during follow-up were associated with the occurrence of at least 1 seizure

Table 6. Outcome for Specific ILAE Epileptic Syndromes

	Frequency	Percentage
Idiopathic generalized epilepsy syndrome	39	
Juvenile absence epilepsy	2	
Needed at least 2 AED to achieve seizure control	0	0
Remained on antiepileptic medication at last visit	0	0
Excellent seizure control in prior year	2	100
Disposition at last visit		
Discharged from neurology	2	100
Continue care with pediatric neurology	0	0
Transferred to adult neurology	0	0
Juvenile myoclonic epilepsy	13	
Needed at least 2 AED to achieve seizure control	3	23.1
Remained on antiepileptic medication at last visit	11	84.6
Excellent seizure control in prior year	11	84.6
Disposition at last visit		
Discharged from neurology	2	15.4
Continue care with pediatric neurology	2	15.4
Transferred to adult neurology	9	69.2
Other generalized idiopathic epilepsies ^a	24	
Needed at least 2 AED to achieve seizure control	3	12.5
Remained on antiepileptic medication at last visit	11	45.8
Excellent seizure control in prior year	18	75.0
Disposition at last visit		
Discharged from neurology	13	56.5
Continue care with pediatric neurology	2	8.7
Transferred to adult neurology	8	34.8
Focal idiopathic epilepsy syndrome	4	
Benign epilepsy of childhood with rolandic spikes	1	
Needed at least 2 AED to achieve seizure control	0	0
Remained on antiepileptic medication at last visit	0	0
Excellent seizure control in prior year	1	100
Disposition at last visit		
Discharged from neurology	1	100
Continue care with pediatric neurology	0	0
Transferred to adult neurology	0	0
Benign occipital epilepsy	3	
Needed at least 2 AED to achieve seizure control	0	0
Remained on antiepileptic medication at last visit	2	66.7
Excellent seizure control in prior year	2	66.7
Disposition at last visit		
Discharged from neurology	1	33.3
Continue care with pediatric neurology	1	33.3
Transferred to adult neurology	1	33.3
Cryptogenic focal epilepsy	16	
Needed at least 2 AED to achieve seizure control	2	12.5
Remained on antiepileptic medication at last visit	9	56.3

(continued)

Excellent seizure control in prior year	13	81.3
Disposition at last visit		
Discharged from neurology	7	43.8
Continue care with pediatric neurology	1	6.3
Transferred to adult neurology	8	50.5
Symptomatic focal epilepsy	6	
Needed at least two AED to achieve seizure control	2	33.3
Remained on antiepileptic medication at last visit	3	50.0
Excellent seizure control in prior year	6	100
Disposition at last visit		
Discharged from neurology	3	50.0
Continue care with pediatric neurology	0	0
Transferred to adult neurology	3	50.0

Note: AED, anti-epileptic drug; ILAE, International League Against Epilepsy.

a. Outcome available for 23 patients.

while on medication during the year preceding the last clinic visit. The presence of a compliance issue was also associated with the need of antiepileptic medication at last visit. Noncompliance with treatment is a well-known problem encountered in the adolescent population, and multiple factors can contribute to this weakening of compliance. Adolescents with epilepsy may feel that taking the medication is "stigmatizing" and could represent an obstacle to an identification to their peer group. In addition, normal adolescent behavior, such as defiance, a search for independence, provocation, and challenge of authority, could also lead to noncompliance with medication.^{13,14} In a previous study, one third (33%) of teenagers with epilepsy were found to have demonstrated a poor compliance with their treatment, 43% had a satisfactory degree of compliance, and only 24% had a good degree of compliance.¹⁵ This represents a higher proportion of noncompliance than the one found in our study (18.5%), which could be explained by our identification of a compliance issue based on documentation in the chart by the physician (through questioning or sub-therapeutic serum drug levels) at the time of visit, which could have underestimated the real proportion of noncompliance. On top of the higher risk of seizure recurrence, poor compliance in adolescents has also been found to be associated with restriction of independence in daily life, a lack of harmony in family life, and low self-esteem.¹⁶

Almost 56% of patients remained on antiepileptic medication at last visit. A diagnosis of juvenile myoclonic epilepsy, the presence of coexisting seizures, coexisting myoclonic seizures, age ≤ 14.5 years at first seizure, and the presence of a compliance issue were all associated with the need for antiepileptic medication at last visit. Juvenile myoclonic epilepsy, one of the idiopathic generalized epilepsy syndromes, is a well-described syndrome that usually has its onset during adolescence. The natural history of this disorder is usually one necessitating long-term treatment with antiepileptic medication because of a high

rate of recurrence of seizures after the discontinuation of treatment. In the current study, among the factors associated with the need for antiepileptic medication at last visit, the presence of coexisting myoclonic seizures can easily be linked with a diagnosis of juvenile myoclonic epilepsy. Furthermore, more than half of our patients who had coexisting seizures also had a diagnosis of juvenile myoclonic epilepsy. To our knowledge, this is the first study showing a link between the age at first suspected seizure and a need for prolonged treatment with antiepileptic medication.

Sixty percent of our cohort had a diagnosis of an idiopathic generalized epilepsy syndrome with the remainder diagnosed with a partial epilepsy syndrome. Other studies performed in adolescents with both childhood-onset and adolescent-onset epilepsy have shown similar results with 58% of the cohort with a diagnosis of idiopathic generalized epilepsy,³ whereas other studies have found a smaller proportion of 43.9%.² Moreover, in our study, one third of the adolescents with an idiopathic generalized epilepsy syndrome were classified as having juvenile myoclonic epilepsy and almost 62% were classified with another generalized idiopathic epilepsy syndrome. Previous studies have shown that among patients with idiopathic generalized epilepsy, 1 in 4 to 1 in 5 were diagnosed with juvenile myoclonic epilepsy,¹⁷ which represents a smaller proportion than in our population. Interestingly, no patient was diagnosed with epilepsy with grand mal on awakening in our study. Among the broad group of idiopathic generalized epilepsy, the heterogeneous partition of epileptic syndromes found in different studies could be the result of the overlapping definitions of the individual syndromes. Furthermore, some experts now consider that the individual syndromes included in the idiopathic generalized epilepsies are part of a clinical continuum rather than actually being separate discrete syndromes or disease entities.⁶

A high proportion of learning disorders (15.2%) and mental retardation (6.2%) was found in our sample of patients with adolescent-onset epilepsy. According to the *Diagnostic and Statistical Manual of Mental Disorders* (Fourth Edition, Text Revision), the prevalence of learning disorders in the general population is thought to be 2% to 10%. It is well-known that pediatric epilepsies are associated with a higher rate of learning disorders than in the general population and that up to 50% of children with epilepsy have school-related difficulties.¹⁸ Adolescent-onset epilepsy was traditionally thought to occur in previously normal children without neurologic or neurodevelopmental defects.⁵ At this point, the relationship between adolescent-onset epilepsy and cognition remains largely unknown and needs to be clarified. In fact, some studies showed that teenagers with epilepsy and a comorbid learning disorder are at higher risk of adopting poor health behaviors and thus represent a more vulnerable subset of patients with adolescent-onset epilepsy.¹⁹

In conclusion, a good outcome can generally be expected in adolescent-onset epilepsy. This study showed that a diagnosis of juvenile myoclonic epilepsy and the presence of coexisting seizures, particularly myoclonic seizures, were associated with a prolonged need for antiepileptic medication, which is in agreement with the well-described long-term outcome of this syndrome. In addition, we found a higher rate of learning disorders in our sample than what is described for the general population, which is in opposition to the traditional thought that adolescent-onset epilepsy occurs in "neurologically normal" individuals. Finally, the presence of a compliance issue was found not to surprisingly have a negative impact on seizure control. New strategies need to be developed to improve compliance in this challenging population. This could be started by giving more information on the importance of seizure control or using more concrete arguments such as the possible driving restrictions that need to be imposed in cases of inadequate seizure control.

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